Speech-To-Text and NLP: Comparing De Gasperi and Meloni's Rhetoric

MARCO BELLÒ, University of Padua, Italy

This report compares the rhetoric of Alcide De Gasperi and Giorgia Meloni. After building a corpus of public speeches transcribed with Whisper from Giorgia Meloni's videos on YouTube, I compare them with a corpus of speeches from the Italian Republic's first Prime Minister Alcide De Gasperi. The results show that De Gasperi's speeches are, on average, 30% more diversified in their lexicon, and the corpora have a Jaccard similarity index of 32%. The highest-frequency words in common show transversal themes, such as freedom, peace and nation, while the highest-frequency words not in common reflect the PMs' different political and societal landscapes. I also performed a sentiment analysis on each speech using a BERT classifier. The results underscore the importance of domain-appropriate training data: the model was trained on product reviews, not on political rhetoric, which translated to an average confidence of roughly 40% per classification.

CCS Concepts: • **Computing methodologies → Natural language processing**; **Speech recognition**; • **Social and professional topics → Political speech**; • **Information systems → Sentiment analysis**.

Additional Key Words and Phrases: AI, NLP, Politics, Speeches

1 INTRODUCTION

In traditional photography, image acquisition is a direct process: a physical lens focuses light reflected from an object onto a photosensitive sensor, comprised by billions of photosites each of which records the raw intensity values of light in three channels: Red, Green, and Blue. The post-processing needed to then obtain a polished-looking picture are quite straightforward, since the object of interest is the *external* appearance of the subject.

On the other hand, many modern scientific and medical apparatuses require looking inside an object to capture phenomena where direct visual access is not possibile, for example when taking an X-Ray of a patient. In this paradigm, the raw hardware measurements are often highly degraded, indirect, or completely unrecognizable, and the final image must be computationally reconstructed. This constraint gave rise to the interdisciplinary field of *Computational Imaging* (CI), which leverages machine learning and other paradigms to overcome hardware limitations and reconstruct high-quality images from raw, unstructured sensor data. This represents a paradigm shift in visual and diagnostic modalities, transitioning to an integrated approach that couples physical measurement devices with rigorous data-driven and variational algorithms.

An example of an application of computational imaging is MRI undersampling: the idea is to purposely acquire only a fraction of the raw data to speed-up the process of a factor of 2X to 10X, to minimize claustrophobia in patients. The image is then reconstructed from the incomplete data.

Mathematically, the core of Computational Imaging lies in the distinction between the *forward problem* and the *inverse problem*. The forward problem models the underlying physics of the imaging system. Assuming the true, continuous underlying object is represented by u, and the discrete, noisy measurements recorded by the hardware are y, the acquisition process can be formulated as:

$$y = Au + \eta \tag{1}$$

where A represents the forward operator (e.g., a blurring kernel, a Fourier transform, or a Radon transform) dictated by the system's physics, and η denotes noise added by the measurement. The objective of Computational Imaging is to

Author's address: Marco Bellò, marco.bello.vi@gmail.com, University of Padua, Padua, Italy.

solve the corresponding *inverse problem*: given the corrupted measurements y and the known or estimated forward operator A, find an approximation u' that accurately recovers the true state of the object.

Two concrete and complementary archetypes of such inverse problems are explored in this essay. The first is *Image Scanning Microscopy* (ISM), a fluorescence imaging modality in which a two-dimensional detector array replaces the single-element detector of a confocal microscope, yielding a four-dimensional dataset. Here the forward operator is a convolution with a Point Spread Function (PSF), and the reconstruction task consists of recovering a fluorophore map from multiple noisy and shifted views of the same subject. The second is *Computed Tomography* (CT), where a scanner measures the attenuation of X-ray beams passing through the body at various angles, producing raw data known as a sinogram. In this case the forward operator is the Radon transform, and the reconstruction task is to recover a cross-sectional image from a series of indirect projections. Despite their different physical origins and scales, both problems are governed by the same mathematical structure: a linear forward operator whose inversion is ill-posed, requiring principled regularisation or learned priors to yield stable and high-quality reconstructions.

The last half of this essay explores practical approaches to Computed Tomography, implemented using the *DeepInverse* library. First, we will reconstruct a precise internal cross-sections, with the raw data represented as a sonogram, consisting of discrete, noisy line integrals corrupted by noise. The objective is to formulate a rigorous forward model of the scanning process based on the discrete Radon transform and solve the resulting ill-posed inverse problem to recover the tissue attenuation map. We will implement optimization algorithms for Maximum Likelihood and Maximum a Posteriori estimation.

Then, we will try to learn the prior distribution implicitly from data. We will explore *Plug-and-Play (PnP) Priors*, replacing the proximal operator of the regularizer with a pre-trained deep denoising neural network. This is a hybrid approach that combines known physics with learned image statistics. Lastly, we will train our own denoiser to use it in a PGD network.

Sources. The main sources for this essay are the following:

- Prof. Luca Calatroni's mini-course slides "Computational imaging & learning: from physics to data-driven methods";
- C. A. Bouman, Foundations of Computational Imaging: A Model-Based Approach, SIAM, 2022 [2];
- G. Aubert, P. Kornprobst, Mathematical Problems in Image Processing: PDEs and the COV, II edition, Springer, 2000 [1];
- Tony F. Chan, J. Shen, Image Processing and Analysis: variational, PDE, wavelet and stochastic methods, SIAM, 2005 [3];
- Tristan van Leeuwen's blog [7];
- Johannes Meyer's blog [5];
- Numerical tours by G. Peyré [6];
- DeepInverse tutorials [4].

2 IMAGE SCANNING MICROSCOPY

Modern biological and biomedical research frequently demands the ability to image structures at the nanoscale — from individual proteins a few nanometres in size, to organelles and whole cells at the micrometre scale. Optical microscopy has long been the workhorse of such investigations, owing to its non-invasive nature and compatibility with live specimens. Yet imaging at this scale is far from straightforward: two fundamental requirements, *high contrast* and *high resolution*, are each difficult to achieve and often stand in tension with one another.

2.1 Fluorescence Microscopy and the Diffraction Limit

Fluorescence microscopy addresses the contrast challenge by exploiting the phenomenon of fluorescence: certain molecules (fluorophores) absorb excitation light at a shorter wavelength and re-emit it at a longer, less energetic wavelength. By selectively labelling specific cellular structures with fluorescent markers, one can visualise components of interest against a dark background with high specificity, effectively filtering out the rest of the sample. A standard fluorescence microscope routes a laser excitation beam through an objective lens onto the specimen; the emitted fluorescent light is then separated from the excitation by a dichroic mirror and emission filter, and collected by a detector to form an image.

Resolution, however, is fundamentally bounded by the wave nature of light. A lens cannot focus an electromagnetic beam to an infinitely small spot; instead, each point source in the sample is imaged as a diffuse blob known as the *Point Spread Function* (PSF). The width of the PSF determines the smallest resolvable feature: two emitters closer together than the PSF width are indistinguishable. This is formalised by the *Abbe diffraction limit*, which states that the minimal resolvable distance d satisfies

$$d = \frac{\lambda}{2\,\mathrm{NA}} = \frac{\lambda}{2n\sin\theta} \approx 200\text{–}250\ \mathrm{nm}, \tag{2}$$

where λ is the emission wavelength, $\mathrm{NA} = n\sin\theta$ is the numerical aperture of the objective, n is the refractive index of the immersion medium, and θ is the half-angle of the acceptance cone. Equivalently, in the frequency domain, the *Optical Transfer Function* (OTF) — the Fourier transform of the PSF — acts as a low-pass filter, irreversibly suppressing spatial frequencies beyond a cutoff determined by the numerical aperture. The consequence is that widefield fluorescence microscopy, despite its excellent contrast, is resolution-limited to roughly a quarter of a micrometre, leaving many sub-cellular structures inaccessible.

2.2 Confocal Laser Scanning Microscopy

The *Confocal Laser Scanning Microscope* (CLSM) improves upon widefield imaging by replacing camera-based wide-area detection with a point-by-point scanning strategy. A focused laser beam is raster-scanned across the specimen using galvanometric mirrors; at each scan position $\mathbf{x}_s$, the emitted fluorescence passes back through the objective and is focused through a small *pinhole* aperture onto a single-element detector (typically a photodiode or avalanche photodetector), which records a single photon-count value. The pinhole physically rejects out-of-focus light, conferring two important advantages: optical sectioning (the ability to reconstruct thin focal planes within a thick specimen) and, when the pinhole is closed down, a modest improvement in lateral resolution beyond the Abbe limit by up to a factor of $\sqrt{2}$.

These gains come at a cost. Closing the pinhole necessarily discards a large fraction of the collected photons, severely reducing the signal-to-noise ratio (SNR). The experimenter thus faces a hard trade-off: a wide pinhole recovers photon counts but degrades both sectioning and resolution, while a narrow pinhole improves spatial quality at the expense of a dim, noisy image. This photon-budget constraint is particularly problematic in live-cell imaging, where fluorophore concentrations and permissible laser powers are both limited by phototoxicity.

2.3 Image Scanning Microscopy: Concept and Data Structure

Image Scanning Microscopy (ISM) resolves the CLSM trade-off by replacing the single-element detector with a two-dimensional *Single-Photon Avalanche Diode (SPAD) array*. The array, subtending approximately 1.4 Airy Units at the detector plane, is centred on the optical axis. For each scan position $\mathbf{x}_s = (x_s, y_s)$, rather than integrating all arriving photons into one number, every pixel $\mathbf{x}_d = (x_d, y_d)$ of the detector array records its own photon count independently. A complete ISM acquisition therefore produces a *four-dimensional dataset* $i(\mathbf{x}_s \mid \mathbf{x}_d)$, consisting of $(N_x \times N_y)$ micro-images (one per scan point), each of which is a small image of the sample as seen by a particular detector element. Equivalently, one may reorganise the same data as N_{ch} scanned images (one per detector pixel), each representing a full field of view acquired from a different vantage point on the detector plane. This high-dimensional dataset carries substantially more spatial information than a conventional confocal acquisition and is a defining feature of the ISM modality.

2.4 Forward Model and the Inverse Problem

The image formation process in fluorescence microscopy is well described as a *linear shift-invariant* (LSI) system. The observed image is the convolution of the unknown fluorophore distribution $\mathbf{u} \in \mathbb{R}_+^N$ with the system's PSF h, corrupted by Poisson photon-counting noise $\mathcal{P}$:

$$\mathbf{y} = \mathcal{P}(\mathbf{h} * \mathbf{u}) = \mathcal{P}(\mathbf{A}\mathbf{u}), \tag{3}$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the Toeplitz matrix encoding convolution with the PSF. In CLSM the PSF is a product of the excitation PSF and the effective detection PSF shaped by the pinhole; in the ISM setting, the forward model generalises to a family of N_{ch} equations

$$\mathbf{y}_d = \mathcal{P}(\mathbf{h}_d * \mathbf{u}) = \mathcal{P}(\mathbf{A}_d \mathbf{u}), \qquad d = 1, \ldots, N_{\text{ch}}, \tag{4}$$

where each detector element $\mathbf{x}_d$ produces a scanned image governed by a *distinct* PSF $h_d(\mathbf{x}_s \mid \mathbf{x}_d) = h_{\text{exc}}(\mathbf{x}_s) \times h_{\text{det}}(\mathbf{x}_s - \mathbf{x}_d)$. The effective PSF for detector element d can be approximated as a shifted copy of a common PSF, $h(\mathbf{x}_s \mid \mathbf{x}_d) \approx h(\mathbf{x}_s - \boldsymbol{\mu}(\mathbf{x}_d))$, where $\boldsymbol{\mu}(\mathbf{x}_d)$ is the *shift-vector* associated with that detector element — computed as the shift maximising cross-correlation between micro-images. Because each detector element occupies a different position relative to the optical axis, the N_{ch} scanned images represent shifted, hence *redundant*, views of the same underlying object. This redundancy is the key resource that ISM exploits to surpass the confocal resolution limit while recovering the photon budget lost by closing the pinhole. The SPAD array introduces an additional practical advantage: it produces no electronic read-out noise, so photon statistics are governed purely by Poisson shot noise — a clean statistical model amenable to principled computational treatment.

Recovering the fluorophore distribution $\mathbf{u}$ from the measured ISM dataset $\{i(\mathbf{x}_s \mid \mathbf{x}_d)\}$ is thus an *inverse problem*: given the forward operators $\{\mathbf{A}_d\}$ and the noisy observations, infer the latent image. Naive approaches such as pixel-reassignment (summing the shifted micro-images) provide a fast but suboptimal estimate; modern reconstruction pipelines exploit the full 4D dataset through variational regularisation or learned priors to achieve simultaneous super-resolution and optical sectioning.

2.5 From Microscopy to Tomography

The inverse problem structure of ISM shares deep mathematical parallels with that of Computed Tomography, discussed in the following section. In both cases, one seeks to recover an unknown spatial distribution — a fluorophore map $\mathbf{u}(\mathbf{x})$ in ISM, an attenuation map $\mu(\mathbf{x})$ in CT — from a collection of indirect, noisy measurements related to it through a linear forward operator. In ISM the forward operator is a convolution (a consequence of the LSI imaging model), while in CT it is the Radon transform (a consequence of straight-line X-ray propagation); yet both are compact linear operators whose inversion is ill-posed and requires regularisation. Moreover, just as ISM acquires multiple shifted views of the same object — each providing complementary spatial information — CT acquires projections from multiple angles, with each angular view illuminating a different set of line integrals through the target. The transition from microscopy to tomography is therefore not merely a change of application domain, but a shift in the geometry of the forward operator: from local blurring in the spatial domain to global integration along lines. Both settings exemplify the broader programme of computational imaging, in which the design of the acquisition, the modelling of the forward physics, and the inversion algorithm are co-optimised to extract the maximum information about an object from measurements that are always finite, incomplete, and noisy.

3 COMPUTED TOMOGRAPHY

Computed Tomography (CT) is a non-invasive imaging technique that reconstructs the internal structure of an object from measurements taken externally. The term *tomography* itself derives from the Greek words *tomos* (slice or section) and *graphein* (to write or record), which reflect its purpose of producing representations of a target by observing

cross-sectional images of a target without physical intervention [**?**].

The physical principle underlying CT relies on the transmission of waves or particles, most commonly X-rays, through the object of interest. As X-rays traverse matter, their intensity is attenuated according to the material's local properties. In the simplest homogeneous case, this relationship is captured by the exponential decay $I_1 = I_0 e^{-\mu s}$, where I_0 and I_1 denote the incident and transmitted intensities, $\mu > 0$ is the attenuation coefficient, and s is the path length through the material. For non-homogeneous objects, this generalises to the *Beer–Lambert law*:

$$I_1 = I_0 \, e^{-\int_\ell \mu(x)\, dx}, \tag{5}$$

which, upon a logarithmic transformation, yields the line integral relation $-\log(I_1/I_0) = \int_\ell \mu(x)\, dx$. This linearised measurement, referred to as the *absorption* or *projection*, forms the basis of the tomographic inverse problem. A collection of such projections acquired over multiple angles constitutes a *sinogram*.

Mathematically, CT reconstruction can be cast as an inverse problem. Given a target $u \in X = L^2(\Omega)$ with $u\colon \mathbb{R}^d \to \mathbb{R}_+$ defined on a bounded domain $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$), the *forward problem* models the acquisition process via the Radon transform $\mathcal{R}(u) = y$, which integrates u over lines $\ell(\omega, s)$ parameterised by angle $\omega \in S^1$ and offset $s \in \mathbb{R}$. The reconstruction task, i.e., recovering u from measured data y, constitutes the corresponding *inverse problem*.

The classical analytic solution to this inverse problem is *Filtered Backprojection* (FBP), an exact inversion formula valid in the continuous setting. FBP performs well when complete projection data are available and the target can be assumed static. However, it suffers from theoretical limitations under discretisation and provides no stability guarantees in realistic acquisition conditions.

In many practical scenarios, reducing the X-ray radiation dose or shortening scan time leads to *limited data tomography*, which encompasses two main regimes: *limited-angle CT*, where projections are acquired only within a restricted angular range, and *sparse-angle CT*, where a reduced number of projection angles are sampled over the full range. Both settings cause FBP to produce severely degraded reconstructions, motivating the development of more advanced reconstruction methods capable of handling incomplete and noisy measurement data.

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